MILITARY SPECIFICATION

DEXTRIN

(FOR BIOLOGICAL USE)

1. SCOPE

1.1 This specification covers one type of dextrin for use as a stabilizer in biological culture media.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.- The following specifications, of the issue in effect on the date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

94-40645 - Marking; Exterior, Domestic and Export Shipment, by Contractors.
100-2 - Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

(Army.- Copies of specifications may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Navy.- Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N.Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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(Air Force.- Copies of Federal, Military, and U. S. Army specifications may be obtained upon application to the Commanding General, Air Development Force, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following publications, of the issue in effect on the date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N.Y.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and other Dangerous Articles, etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. may be obtained from the Interstate Commerce Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Appearance.- The dextrin shall be a free flowing white powder.

3.2 Solubility in water.- The material shall contain not more than 1 percent of water-insoluble matter when tested as specified in 4.4.3.

3.3 Solubility in ethanol.- The material shall contain not more than 1 percent of alcohol-soluble matter when tested as specified in 4.4.4.

3.4 Residue upon ignition.- The material shall leave not more than 0.5 percent of residue upon ignition when tested as specified in 4.4.5.

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3.5 Heavy metals.- The material shall contain not more than 0.001 percent of heavy metals when tested as specified in 4.4.6.

3.6 Reducing sugars.- The material shall contain not more than 5 percent of reducing sugars as dextrose when tested as specified in 4.4.7.

3.7 Moisture content (uncombined).- The material shall contain not more than 12 percent of moisture when tested as specified in 4.4.8.

3.8 Oxalate content,- The material shall contain not more than 0.05 percent of oxalate when tested as specified in 4.4.9.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- A lot shall consist of the material produced by one manufacturer with no change in process or materials in no more than 24 consecutive hours, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.2 Sampling.- From each lot five containers shall be taken at random and from each container a 1-pound specimen shall be taken. If there are fewer than five containers, a 1-pound specimen shall be taken from each container (see 6.3).

4.3 Inspection.

4.3.1 Packing and marking.- The inspector shall inspect the packing and marking of the material for conformance to section 5.

4.3.2 Procedures.- For Naval purchases, general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Tests.- All specimens selected in accordance with 4.2 shall be examined for compliance with the heavy metals test specified in 4.4.6. Inspection may be stopped if a failure is observed. If all specimens conform to the heavy metals requirement, then three specimens shall be analysed in accordance with the following procedures:

4.4.1 Reagents.- Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the tests, and corrections shall be applied when significant.

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4.4.2 Appearance.- Visually examine the material for compliance with 3.1.

4.4.3 Solubility in water.- Weigh, to the nearest mg., approximately 10 gm. of the material and transfer to a 250-ml. beaker. Add slowly and with constant agitation enough water heated to 60° ± 5°C. to insure complete solution. When solution is effected, filter through a tared Gooch crucible, wash thoroughly with hot water, heat at 103° ± 3°C. for 4 hours, cool, weigh, and calculate the percent water-insoluble matter.

4.4.4 Solubility in ethanol.- Weigh, to the nearest mg., approximately 10 gm. of the material and transfer to a 250-ml. beaker. Add slowly and with constant agitation 150 ml. of 95-percent ethanol. Boil, allow the insoluble material to settle, and then decant the supernatant hot liquid through an appropriate filter. Evaporate the filtrate or an aliquot at 75° ± 2°C., determine the weight of the residue, and calculate the percent alcohol-soluble material.

4.4.5 Residue upon ignition.- Weigh, to the nearest mg., approximately 1 gm. of the material and transfer to a tared platinum or silica evaporating dish. Ignite at 600°C. until the ash is white, cool to room temperature in a desiccator, weigh, and calculate the percent residue (ash) in the sample.

4.4.6 Heavy metals.

4.4.6.1 Test solutions.

4.4.6.1.1 Bromine solution (bromine water).- Prepare a saturated aqueous solution of bromine, by agitating from 2 to 3 ml. of bromine with 100 ml. of cold water in a glass-stoppered bottle, the stopper of which has been lubricated with petrolatum. Preserve in a cool dark place.

4.4.6.1.2 Acetic acid solution.- Prepare an acetic acid solution by diluting 60 ml. of glacial acetic acid to 1 liter with water.

4.4.6.1.3 Ammonia solution.- Prepare an ammonia solution by diluting 395 ml. of concentrated reagent ammonia water (28-percent) to 1 liter with water.

4.4.6.1.4 Hydrogen sulfide solution.- Prepare a saturated aqueous solution of hydrogen sulfide by passing hydrogen sulfide into cold water.

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4.4.6.1.5 Stock solution of lead nitrate.- Dissolve 0.1598 gm. of lead nitrate in 100 ml. of water, to which has been added 1 ml. of nitric acid, and dilute to 1 liter with water.

4.4.6.1.6 Standard lead solution.- Dilute 10 ml. of the stock solution of lead nitrate (see 4.4.6.1.5). to 100 ml. with water. Prepare fresh for each test. Each ml. of this standard lead solution contains the equivalent of 0.01 mg. of lead. When 0.1 ml. of standard lead solution is employed to prepare the standard to be compared with 1 gm. of the substance being tested, the comparison solution thus prepared contains the equivalent of 1 part of lead per million parts of the substance tested.

4.4.6.2 Procedure.

4.4.6.2.1 Solution A.- Mix 2 ml. of diluted acetic acid (see 4.4.6.1.2), 5 ml. of standard lead solution (see 4.4.6.1.6), and dilute to 25 ml. with water.

4.4.6.2.2 Solution B.- Thoroughly ash 5 gm. of the dextrin to be tested in a porcelain dish, cool, add a mixture of 30 ml. of diluted hydrochloric acid (236 ml. conc. hydrochloric acid diluted with water to 1 liter) and 3 ml. of bromine solution (see 4.4.6.1.1); then cover the dish and boil gently for 10 minutes. Filter and wash the filter and residue with 25 ml. of hot water. Evaporate the filtrate to approximately 15 ml., cool, and add ammonia solution (see 4.4.6.1.3), dropwise, until the solution is neutral to litmus paper, then add 2 ml. of acetic acid solution (see 4.4.6.1.2) and dilute to 25 ml. with water.

4.4.6.2.3 Comparison.- Transfer solutions A and B to similar 50-ml. Nessler tubes, add 10 ml. of hydrogen sulfide solution (see 4.4.6.1.4) to each tube, mix, allow to stand for 10 minutes, then view downward over a white surface: When the color of B is no darker than that of A, the heavy metals content of the dextrin tested is not greater than 0.001 percent.

Note.- The presence of heavy metals may be detected qualitatively by dissolving 1 gm. of the dextrin in 10 ml. of water and adding hydrogen sulfide water. Any change indicates the presence of heavy metals and would necessitate following the quantitative procedure contained in 4.4.6.

4.4.7 Reducing sugars.

4.4.7.1 Test solutions.

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solution present at the end of titration and titrate with approximately 0.1N sodium thiosulfate using starch as an indicator. Calculate the percent reducing sugars as dextrose as follows:

$$\text{Percent reducing sugars as dextrose} = \frac{100\ A}{W}$$

$$\text{gm. copper reduced} = (V-v)N \times 0.0636$$

where:

A = gm. of dextrose equivalent to gm. of copper reduced as obtained from tables for the determination of dextrose (see 6.4).

W = weight of sample in aliquot.

V = ml. thiosulfate solution used for test.

v = ml. thiosulfate solution used for blank.

N = normality of thiosulfate solution.

4.4.8 Moisture content (uncombined).- Weigh, to the nearest mg., approximately 5 gm. of the material, transfer to a suitable drying dish, dry for 4 hours at 103° ± 3°C., weigh, and calculate the percent moisture in the sample.

4.4.9 Oxalate content.- Mix approximately 1.600 gm. of the material with 20 ml. of cold water, filter, and add 5 ml. of calcium chloride solution to the filtrate. If a turbidity develops, the oxalate content is greater than 0.05 percent.

4.5 Rejection and resubmission.- If any specimen fails to conform to the requirements of this specification, the lot represented by the nonconforming specimen shall be rejected. The contractor, at no expense to the Government, shall have the option of having a partial or complete analysis made of each container in the rejected lot. He may then remove the nonconforming containers and resubmit the remaining portion of the lot for acceptance testing.

5. PREPARATION FOR DELIVERY

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5.1 Packaging.- The dextrin shall be packed in drums conforming to the requirements of Interstate Commerce Commission Specification 21A. The drums shall be lined with a polyethylene liner having a nominal thickness of 4 mils. The weight of the dextrin in a drum shall not exceed 300 pounds (see 6.1).

5.2 Marking.- In addition to any special marking required by the contract or order, all shipping containers for the Army shall be marked in accordance with Specification 100-2; for the Air Force, in accordance with Specification 94-40645; and for the Navy, in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size of drum (see 5.1).

6.2 Inspection.- Inspection will not necessarily include all drawing and specification requirements; however, the omission of inspection does not constitute a waiver of these requirements.

6.3 Sampling and testing.- When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.4 Tables for dextrose determination.- Tables for the determination of dextrose from reduced copper may be found in the following publications:

a. Official and Tentative Methods of Analysis of the Association of Official Agricultural Chemists.
b. Handbook of Chemistry and Physics - Chemical Rubber Publication, 31st Edition, page 1470.
c. Journal American Chemical Society, Volume 28, page 663, (1906).

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6.5 Batch.- A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Army - Chemical Corps.

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